

LAB EXPERIMENT RECORD

Set 3: Employee Performance Evaluation

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
Name	Swetha S
Register Number	192424049
Date	29 September 2026
Day	Tuesday

Aim

To analyze employee performance data using R programming by creating a line chart, bar chart, and scatter plot, and to develop a Tableau dashboard for interactive exploration of employee performance and department-wise employee distribution.

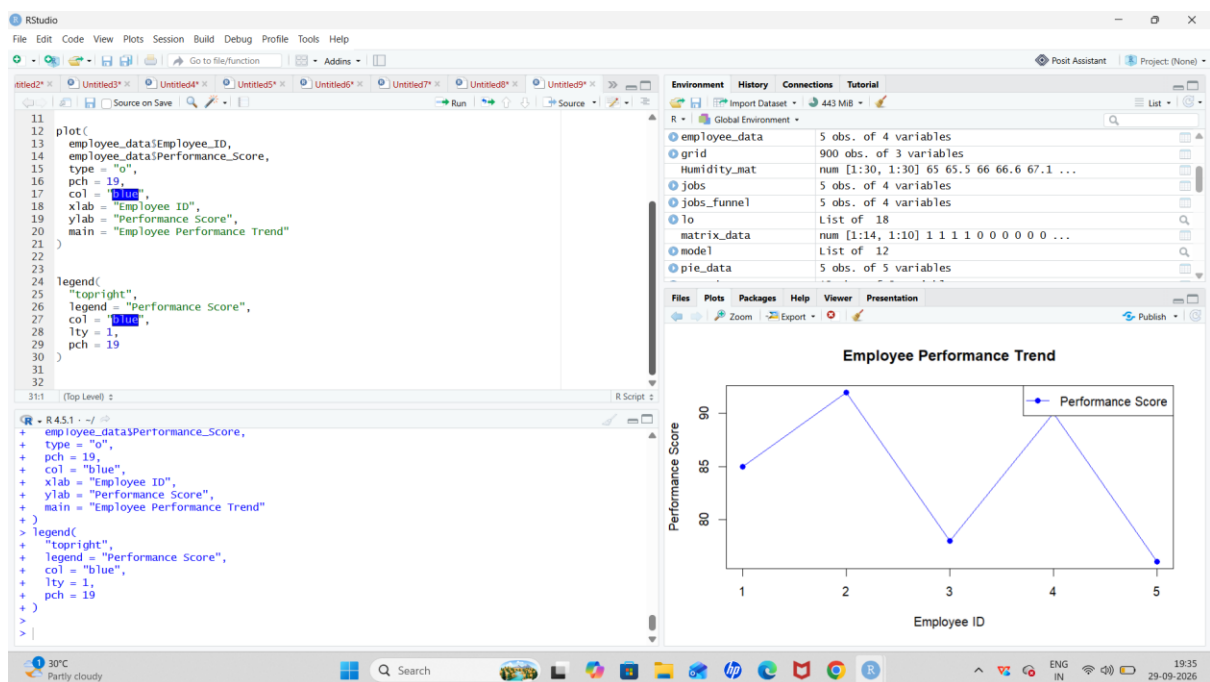
Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

1. Create a line chart to visualize the performance trend of employees over time. Include a legend and Labels.

```
plot(  
  employee_data$Employee_ID,  
  employee_data$Performance_Score,  
  type = "o",  
  pch = 19,  
  col = "blue",  
  xlab = "Employee ID",  
  ylab = "Performance Score",  
  main = "Employee Performance Trend"  
)
```

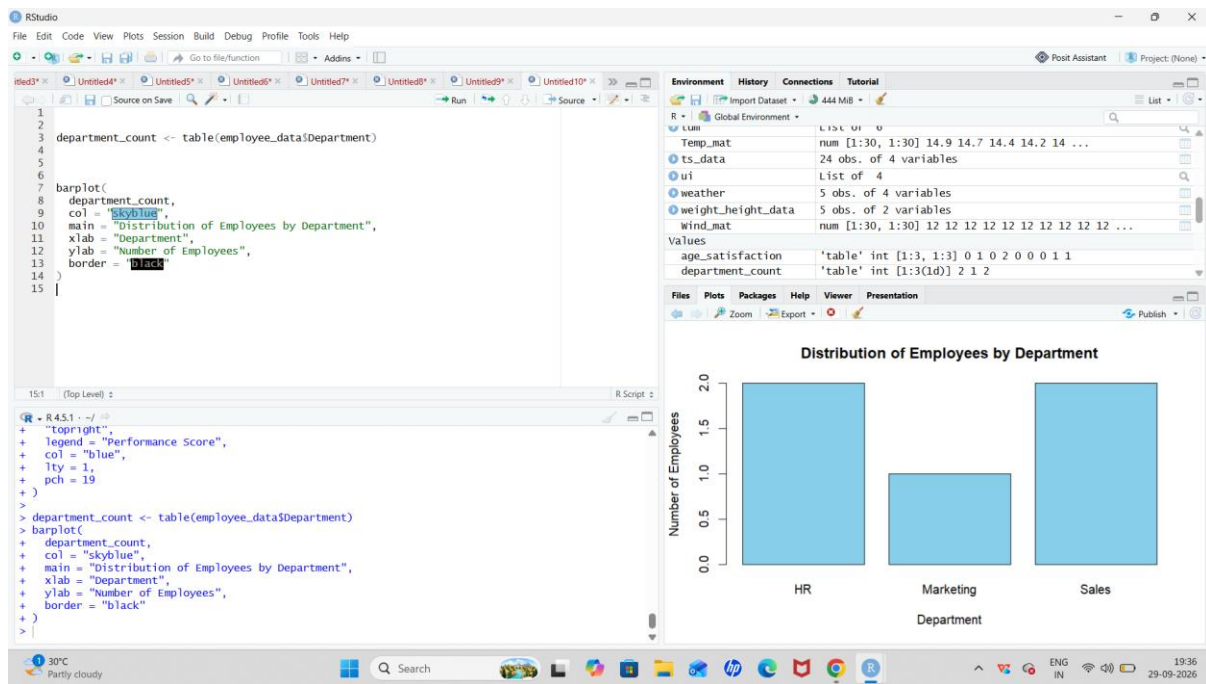
```
legend(  
  "topright",  
  legend = "Performance Score",  
  col = "blue",
```



2. Generate a bar chart showing the distribution of employees across different departments.

```
department_count <- table(employee_data$Department)
```

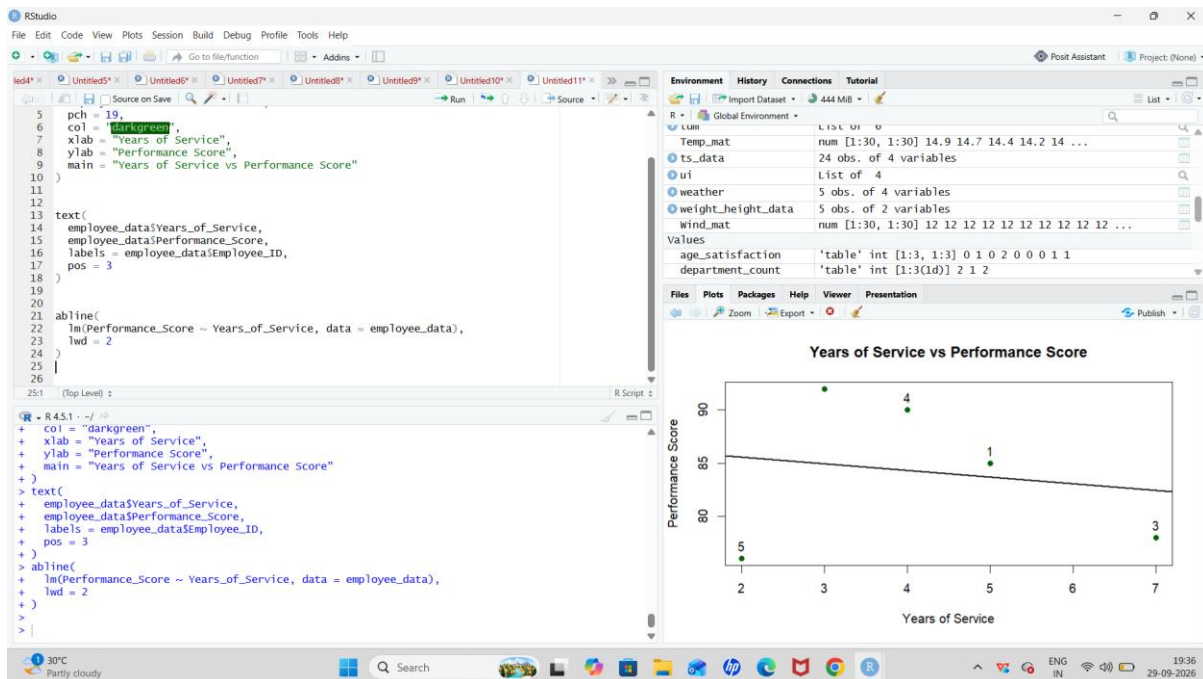
```
barplot(  
  department_count,  
  col = "skyblue",  
  main = "Distribution of Employees by Department",  
  xlab = "Department",  
  ylab = "Number of Employees",  
  border = "black"  
)
```



3. Build a scatter plot to analyze the correlation between years of service and performance scores.

```
plot(
  employee_data$Years_of_Service,
  employee_data$Performance_Score,
  pch = 19,
  col = "darkgreen",
  xlab = "Years of Service",
  ylab = "Performance Score",
  main = "Years of Service vs Performance Score"
)
```

```
text(
  employee_data$Years_of_Service,
  employee_data$Performance_Score,
  labels = employee_data$Employee_ID,
  pos = 3
)
```



4. Develop a Tableau dashboard with the line chart and bar chart for interactive exploration.

Step 1: Enter the data

Create an Excel file with:

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

Open this Excel file in **Tableau**.

Step 2: Create the line chart

In Tableau:

1. Open a new **Worksheet**.
2. Drag **Employee ID** to **Columns**.
3. Drag **Performance Score** to **Rows**.
4. Select **Line** from the Marks card.
5. Turn on **Show Markers** if available.
6. Add **Department** to **Color** if you want departments represented separately.
7. Rename the sheet as **Employee Performance Trend**.

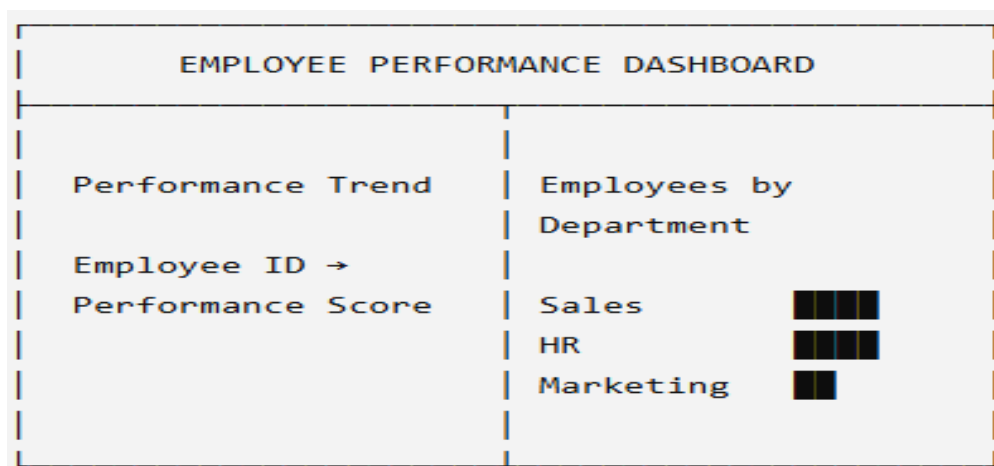
Step 3: Create the bar chart

1. Create another worksheet.
2. Drag **Department** to **Columns**.
3. Drag **Employee ID** to **Rows**.
4. Change Employee ID aggregation to **COUNT**.
5. Select **Bar** from the Marks card.
6. Turn on **Show Mark Labels**.
7. Rename it **Employees by Department**.

Step 4: Create the dashboard

1. Click **New Dashboard**.
2. Drag **Employee Performance Trend** onto the dashboard.
3. Drag **Employees by Department** below or beside it.
4. Add filters such as **Department**.
5. Enable interactions so selecting a department in the bar chart can filter the other chart.

Dashboard structure



Procedure

1. Create the employee performance dataset in R with Employee ID, Department, Years of Service, and Performance Score.
2. Use the plot() function to create a line chart showing the performance scores of employees.
3. Add appropriate axis labels, title, and legend to the line chart.
4. Use the table() function to calculate the number of employees in each department.
5. Use the barplot() function to create a bar chart showing the distribution of employees across departments.
6. Create a scatter plot using years of service and performance scores to analyze their relationship.

7. Add a trend line using the `lm()` function to observe the correlation pattern.
8. Calculate the correlation between years of service and performance score using the `cor()` function.
9. Import the employee dataset into Tableau.
10. Create a line chart for employee performance and a bar chart for department-wise employee distribution.
11. Combine both charts into a Tableau dashboard and add filters for interactive exploration.

Result

The employee performance data was successfully analyzed using R. The line chart displayed the performance scores of employees, while the bar chart showed the number of employees in each department. The scatter plot helped analyze the relationship between years of service and performance scores. The analysis showed that years of service alone does not clearly determine performance in this small dataset. Finally, a Tableau dashboard was developed by combining the charts, allowing users to interactively explore employee performance and department-wise information.